



University of Piraeus

School of Information and Telecommunications Technologies

Department of Digital Systems

Level: Postgraduate Program of Study

Cyber Defense and Digital Forensics

Assignment 1

Supervising Professors: Christoforos Dantoyan and Evangelos Dragonas

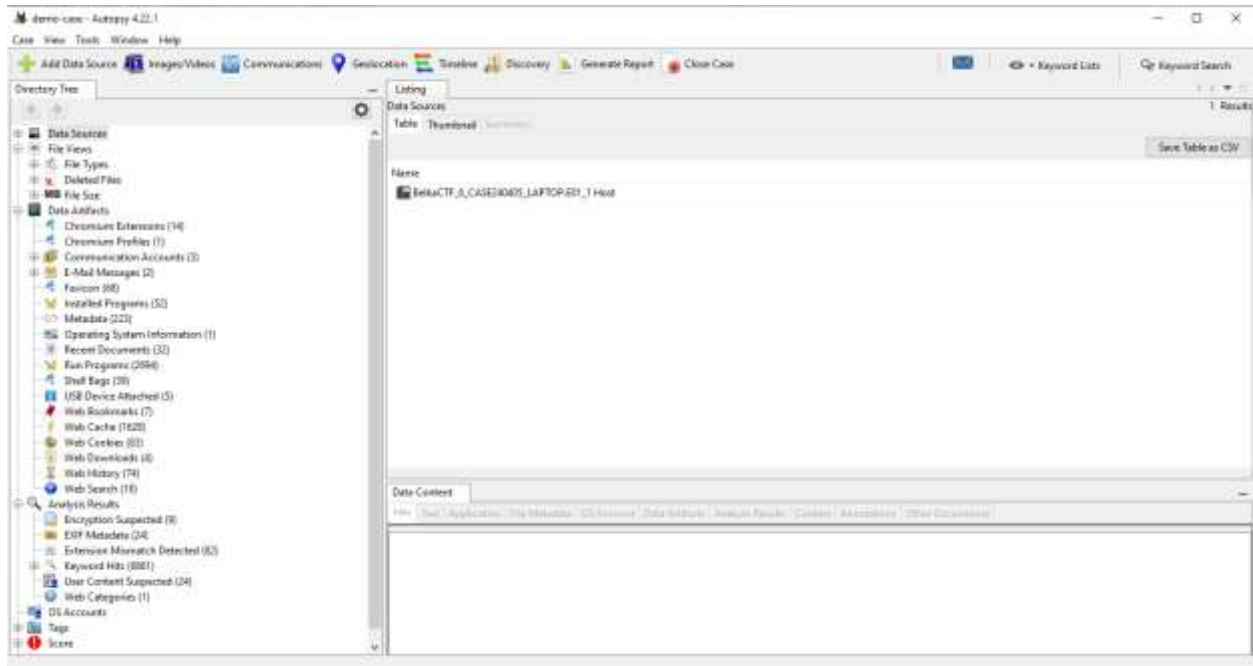
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Piraeus

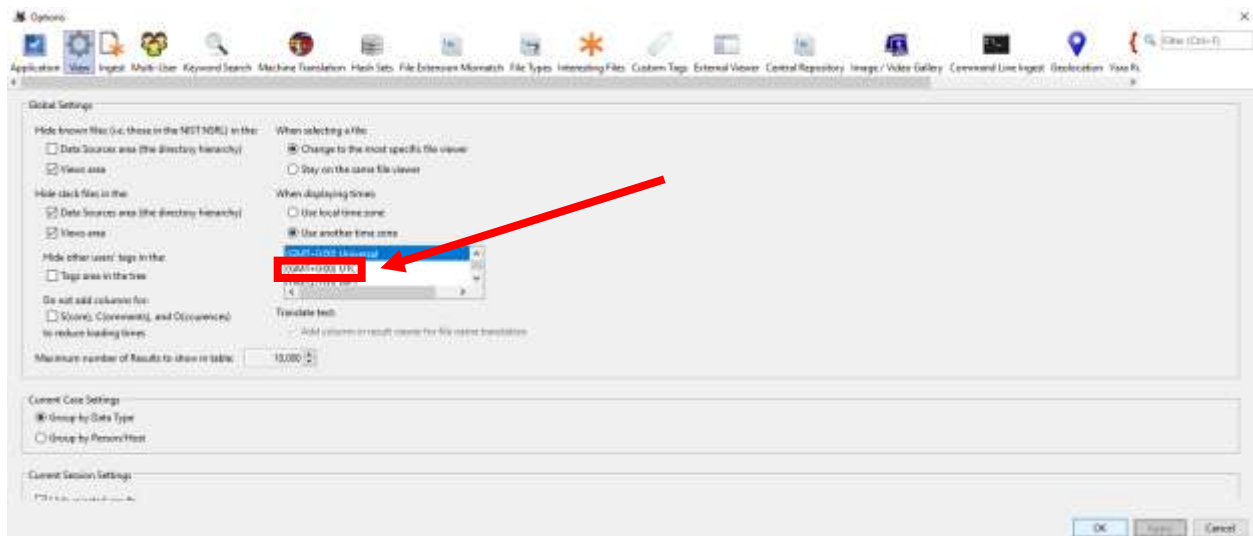
02/04/2026

The prompts for the installation of the Autopsy and the preparation for answering all questions were followed.



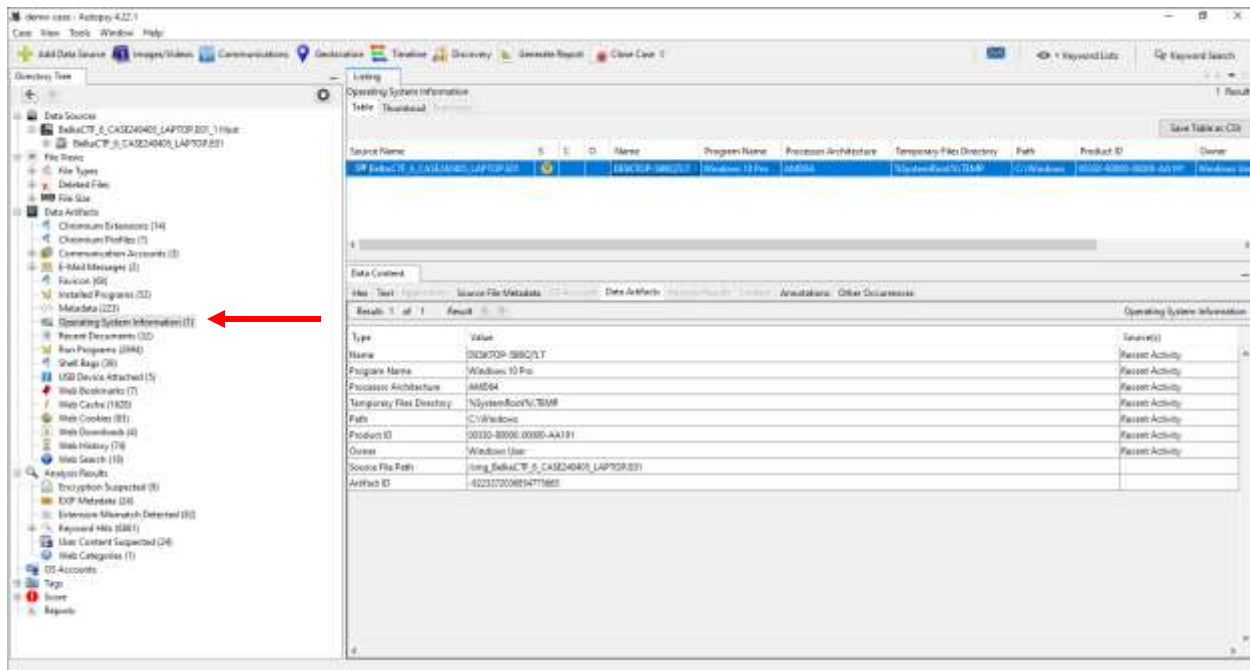
Q1: When was the current version of the operating system installed on this system? (If you used Autopsy to find the answer add a ‘Bookmark’ tag to the artifact that supports your answer- Provide the timestamps in UTC):

Note: If the timezone in Autopsy is not set to UTC, you can change timezone by clicking on Tools->Options->View and change the timezone to UTC. Then click on ‘OK’ to apply.

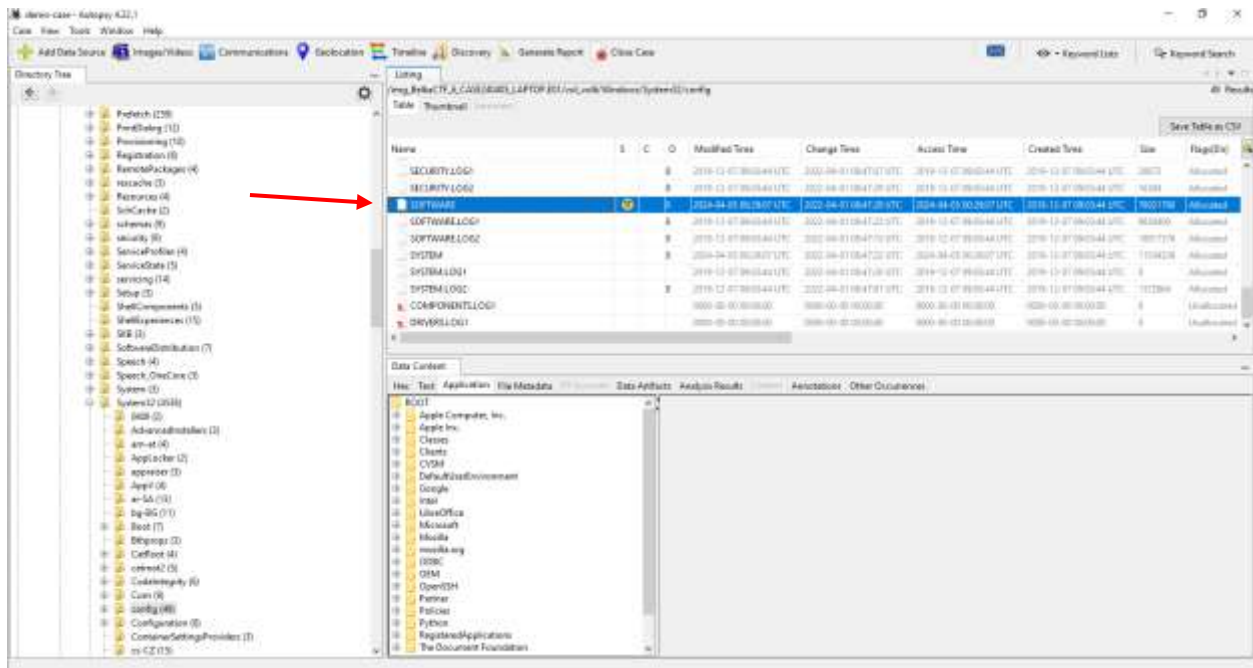


Answer:

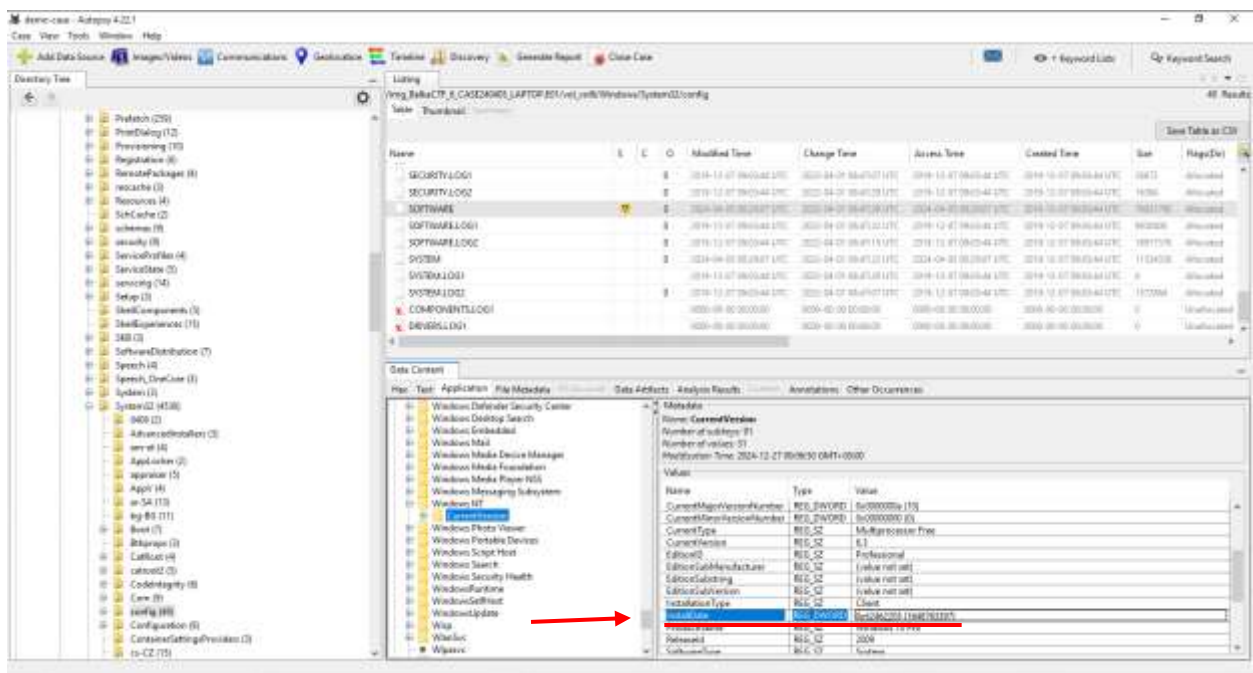
After first setting the correct time zone in Autopsy to UTC (Tools -> Options -> View), we proceeded to find the installation date of the current version.



Initially, the search was carried out in the automated "Operating System Information" section, but the desired result was not detected. Based on the suggestions of the exercise, we proceeded to a manual analysis of the Registry, following the following route: **Data Sources -> BelkaCTF_6_CASE... E01 -> Vol6 (Basic data partition: 67... 71) -> Windows -> System32 -> Config.**

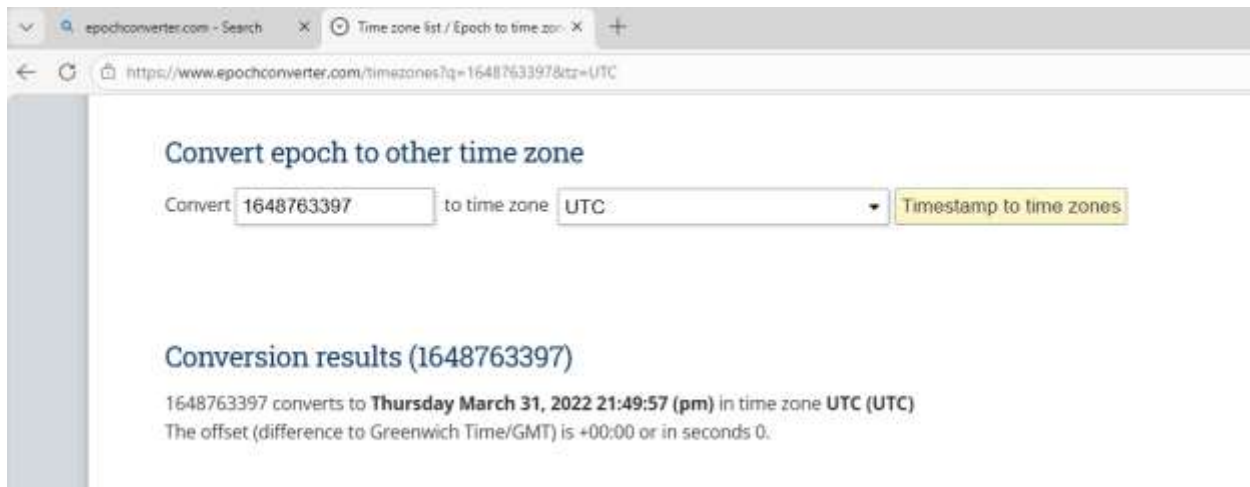


Opening the file «SOFTWARE» in the 'Application' tab, we navigated to **Windows NT -> CurrentVersion**. This is where the 'InstallDate' record was detected, which contained the value **0x62462205 (1648763397)**.



This value is in the form of **Unix Timestamp** (or Epoch time), which represents the number of seconds that have elapsed since January 1, 1970. It's a common way Windows (and other operating systems) store time data in the Registry. To convert this value, the online tool was used

epochconverter.com. By entering the 1648763397 number, it was converted to a UTC time zone.



The OS installation date is **March 31, 2022, at 21:49:57 UTC**.

Q2: Answer the following questions regarding the user with SID “S-1-5-21-1548112901-4150280657-3748420248-1000” (If you used Autopsy to find the answer add a ‘*Bookmark*’ tag to the artifact that supports your answer- Provide the timestamps in UTC):

- a) What is the username of this account?
- b) When was the last login performed?
- c) When was the password last changed?

Answer:

- a) We find the category called "**OS Accounts**" and left click on it. In the top right window (Listing) a list of all Windows accounts will appear.

Listing
11 Results

Table
Thumbnail
Summary

Save Table as CSV

Name	S	C	O	Login Name	Host	Scope
S-1-5-80-956008885-3418522649-1831038044-185329			0		BelkaCTF...	Local
S-1-5-18				SYSTEM	BelkaCTF...	Local
S-1-5-21-1548112901-4150280657-3748420248-1000			0	phorger	BelkaCTF...	Domain
S-1-5-80-3028837079-3186095147-955107200-370196			0		BelkaCTF...	Local
S-1-5-19				LOCAL SERVICE	BelkaCTF...	Local
S-1-5-20				NETWORK SERVICE	BelkaCTF...	Local
S-1-5-21-3933942852-973373972-2766786355-1032			0		BelkaCTF...	Domain
S-1-5-21-1548112901-4150280657-3748420248-504			0	wdagutilityaccount	BelkaCTF...	Domain

Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences

Hex
Text
Application
File Metadata
OS Account

Basic Properties

Login: phorger
Full Name:
Address: S-1-5-21-1548112901-4150280657-3748420248-1000
Type:
Creation Date: 2022-03-31 21:49:50 UTC
Object ID: 429

BelkaCTF_6_CASE240405_LAPTOP.E01_1 Host Details

Last Login: 2024-12-27 02:06:49 EET
Login Count: 15
Password Settings: Password not required
Flag: Normal user account
Home Directory: C:/Users/phorger

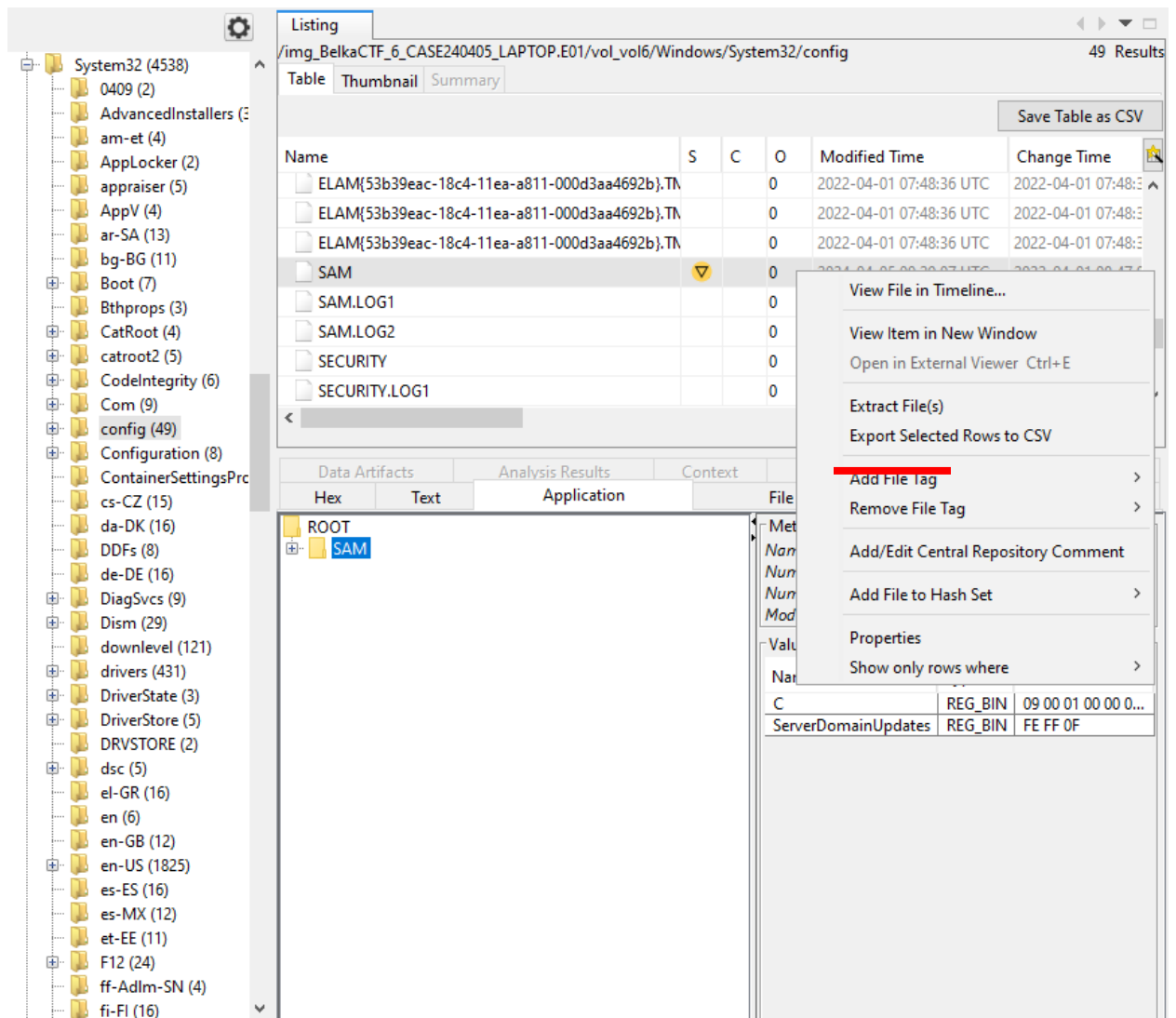
Realm Properties

Name: Unknown
Address: S-1-5-21-1548112901-4150280657-3748420248
Scope: Domain
Confidence: Known

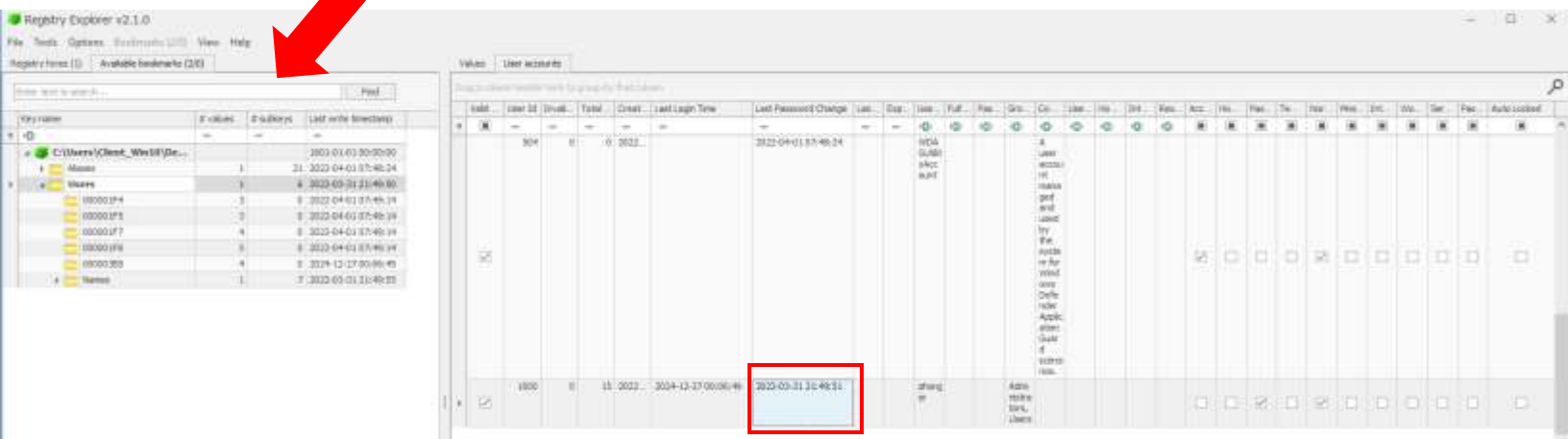
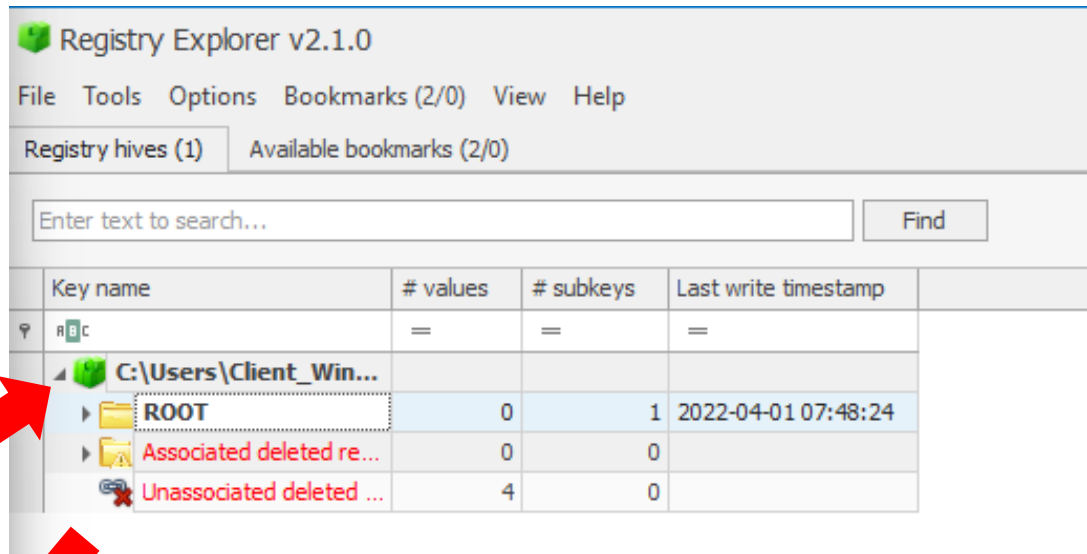
The username is **phorger**.

- b) In the "**Host Details**" section we see the Last Login value: 2024-12-27 02:06:49 EET. For some reason it showed us this value in EET and not in UTC. So the correct answer is: **2024-12-27 00:06:49 UTC**.

- c) In this image, there is no information about the date of the last password change, so we will proceed to use the **Registry Explorer** tool. We return to the path **vol6 -> Windows -> System32 -> config**, in Autopsy and find the **SAM** (Security Account Manager) file.



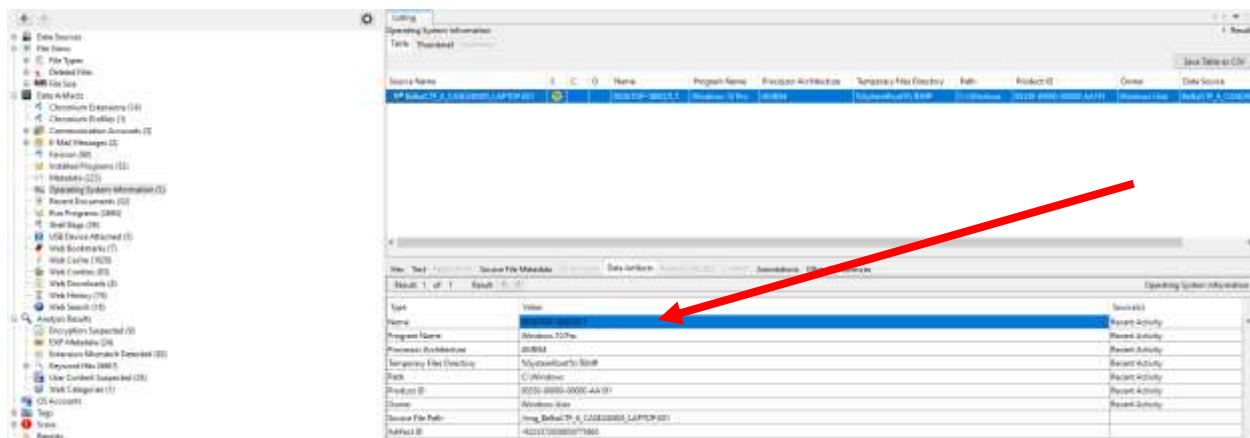
We will do Extract Files on the Desktop and "download" the Registry Explorer. We will put the SAM file in the registry explorer to see the additional information it provides.



For phorger (with User Id 1000), the column "Last Password Change" reads: **2022-03-31 21:49:51**.

Q3: What is the computer name of this system? (If you used Autopsy to find the answer add a 'Bookmark' tag to the artifact that supports your answer)?

Answer:



Within the Autopsy environment, in the left menu (Directory Tree), the subcategory **"Operating System Information»**. This is where the relevant entry was found in both the "Name" column above, and the "Data Artifacts" below, which indicates that the system name is **DESKTOP-5B8Q7LT**.

Q4: Answer the following questions regarding the user *"phorger"*, who searched the web for specific instructions related to anti-forensics (If you used Autopsy to find the answer add a *'Bookmark'* tag to the artifact that supports your answer- Provide the timestamps in UTC):

- What was the search query?
- When did the user perform this search?
- Which web browser did the user use?
- Which search engine did the user use?

Answer:

- Through Autopsy, the "Web Search" subcategory was selected in the left menu, where the search history is recorded. Looking at the records, the search was found in the last line: **"Empty Recycle Bin in Windows"**. Deleting files and emptying the recycle bin is considered a classic anti-forensics technique (the user's attempt to hide or destroy digital

Data Sources

- File Views
- File Types
- Deleted Files
- File Size
- Data Artifacts**
 - Chromium Extensions (14)
 - Chromium Profiles (1)
 - Communication Accounts (3)
 - E-Mail Messages (2)
 - Favicon (34)
 - Installed Programs (32)
 - Metadata (22)
 - Operating System Information (1)
 - Recent Documents (32)
 - Run Programs (294)
 - Shell Bags (3)
 - USB Devices Attached (5)
 - Web Bookmarks (7)
 - Web Caches (152)
 - Web Cookies (31)
 - Web Downloads (4)
 - Web History (18)
 - Web Search (18)
- Analysis Results
 - Encryption Suspected (3)
 - EXIF Metadata (24)
 - Extension Mispatch Detected (32)
 - Keyboard Hits (6981)
 - User Content Suspected (34)
 - Web Categories (1)
- OS Accounts
- Tags
- Score
- Results

Web Search

Table - Thumbnail [Download](#)

Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
History				bing.com	best christmas presents for giftweek	Microsoft Edge	2024-10-29 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	best christmas presents for giftweek	Microsoft Edge	2024-10-28 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	best christmas presents for giftweek	Microsoft Edge	2024-10-29 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	best christmas presents for giftweek	Microsoft Edge	2024-10-28 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	creative gifts for giftweek christmas	Microsoft Edge	2024-10-28 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	creative gifts for giftweek christmas	Microsoft Edge	2024-10-28 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	creative gifts for giftweek christmas	Microsoft Edge	2024-10-28 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	creative gifts for giftweek christmas	Microsoft Edge	2024-10-28 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	creative gifts for giftweek christmas	Microsoft Edge	2024-10-28 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	creative gifts for giftweek christmas	Microsoft Edge	2024-10-28 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	creative gifts for giftweek christmas	Microsoft Edge	2024-10-28 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	creative gifts for giftweek christmas	Microsoft Edge	2024-10-28 22:29:48 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1
History				bing.com	Empty Recycle Bin in Windows	Microsoft Edge	2024-01-03 01:19:15 UTC	BellaCTF_8_CASE240403_LAPTOP-ED1

Has Text Application Source File Metadata OS Account Data Artifacts Analysis Results Comment Annotations Other Discoveries

Results: 85 of 85 Results

Web Search

Item: Empty Recycle Bin in Windows
 Time: 2024-01-03 01:19:15 UTC
 Domain: bing.com
 Program Name: Microsoft Edge

Source

Host: BellaCTF_8_CASE240403_LAPTOP-ED1, 1 Host
 Data Source: BellaCTF_8_CASE240403_LAPTOP-ED1
 File: /img/BellaCTF_8_CASE240403_LAPTOP-ED1/od_vault/Users/phongem/AppData/Local/Microsoft/Edge/User Data/Default/History

The search date is **2024-01-03 01:19:15 UTC**.

The user used the **Microsoft Edge browser**.

The search engine is **bing.com**.

- The user appears to have downloaded a ZIP archive from the Internet. Within this archive, there is one executable file. Can you calculate its SHA-256 hash value?
- The same executable can be found in the user's Desktop folder. Do these two executables have the same SHA-256 hash value?

a) In the first line of the table we clearly see the **powder.zip file** that was downloaded to the **C:\Users\phorger\Downloads** folder. We need to find the executable (.exe) inside this ZIP and calculate its **SHA-256 hash**.

The screenshot displays a forensic analysis tool interface. The top section, titled 'Listing', shows a table of 'Web Downloads' with 4 results. The second row is highlighted, showing a file named 'powder.zip' downloaded from 'https://powderbox.co.uk/Downloads/powder.zip' on '2024-02-02 22:26:37 UTC'. Below this, a 'Listing' section shows a directory tree for '/img_BelkaCTF_6_CASE240405_LAPTOP.E01/vol6/Users/phorger/Downloads/powder.zip'. The file 'Powder.exe' is highlighted. The bottom section, 'File Metadata', provides detailed information about 'Powder.exe', including its name, type, size, and most importantly, its SHA-256 hash: '0c67ea27c37b2b5382a8420ec6f4adf1a777890ef8327a37fbdf4c3e5dfaf181'.

Source Name	S	C	O	Path	URL	Date Accessed	Domain	Username	Program
History			0	C:\Users\phorger\Downloads\powder.zip	https://powderbox.co.uk/Downloads/powder.zip	2024-02-02 22:26:37 UTC	powderbox.co.uk	Default	Microsoft
History			0	C:\Users\phorger\Downloads\python-3.12.2-amd64.exe	https://www.python.org/ftp/python/3.12.2/python-3.12.2-amd64.exe	2024-02-02 22:50:21 UTC	python.org	Default	Microsoft
Capture2 - Copy.PNG.Zone.Identifier				A:\Users\phorger\AppData\Local\Packages\Microsoft.S...					
Capture2.png.Zone.Identifier				A:\Users\phorger\Pictures\Capture2.png					

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location	MT
Powder.exe			0	2024-04-01 18:15:26 UTC	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	35147	Allocated	Allocated	unknown	/img_BelkaCTF_6_CASE240405_LAPTOP.E01/vol6/...	...
...			0	2024-04-01 18:15:26 UTC	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	4096720	Allocated	Allocated	unknown	/img_BelkaCTF_6_CASE240405_LAPTOP.E01/vol6/...	...
...			0	2024-04-01 18:16:00 UTC	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	8131	Allocated	Allocated	unknown	/img_BelkaCTF_6_CASE240405_LAPTOP.E01/vol6/...	...

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Metadata Name: /img_BelkaCTF_6_CASE240405_LAPTOP.E01/vol6/Users/phorger/Downloads/powder.zip/Powder.exe Type: Derived MIME Type: application/x-dosexec Size: 6686720 File Name Allocation: Allocated Metadata Allocation: Allocated Modified: 2024-04-01 18:15:26 UTC Accessed: 0000-00-00 00:00:00 Created: 0000-00-00 00:00:00 Changed: 0000-00-00 00:00:00 MD5: abba583fcde8b0dd3d3754d90992f1f SHA-256: 0c67ea27c37b2b5382a8420ec6f4adf1a777890ef8327a37fbdf4c3e5dfaf181 Hash Lookup Results: UNKNOWN Internal ID: 386215									

The SHA-256 hash of the Powder.exe that was inside the ZIP is:
0c67ea27c37b2b5382a8420ec6f4adf1a777890ef8327a37fbdf4c3e5dfaf181.

- b) Through the Directory Tree in the **vol6\Users\phorger\Desktop** path, the corresponding executable named Powder.exe was detected. Examining its metadata, it was found that the **SHA-256** value is **0c67ea27c37b2b5382a8420ec6f4adf1a777890ef8327a37fbdf4c3e5dfaf181.** In conclusion, the two executables have exactly the same SHA-256 hash value. This proves beyond a doubt that the executable file found on the user's desktop is the exact, identical copy of what was contained in the compressed file they downloaded from the internet. The file came as it was from the original download, without having undergone the slightest

alteration.

The screenshot shows the Autopsy interface. The top pane displays a file listing for the path `/img_BelkaCTF_6_CASE240403_LAPTOP.E01/vol_vols/Users/phantom/Desktop`. The file `Prodeur.exe` is highlighted. The bottom pane shows the metadata for this file.

Name	S	C	D	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
[current folder]				2024-02-02 22:26:53 UTC	2024-02-02 22:26:53 UTC	2024-04-04 11:14:19 UTC	2022-05-31 21:00:02 UTC	380	Allocated	Allocated	unknown	/img_BelkaCTF_6_CASE240403_LAPTOP.E01/vol_vols/Users/phantom/Desktop
[parent folder]				2023-12-24 23:13:13 UTC	2023-12-24 23:13:13 UTC	2023-12-24 23:13:13 UTC	2022-05-31 21:00:02 UTC	256	Allocated	Allocated	unknown	/img_BelkaCTF_6_CASE240403_LAPTOP.E01/vol_vols/Users/phantom/Desktop
desktop.ini				2022-03-23 21:00:04 UTC	2022-03-23 21:00:04 UTC	2024-04-03 00:28:20 UTC	2022-05-31 21:00:04 UTC	252	Allocated	Allocated	unknown	/img_BelkaCTF_6_CASE240403_LAPTOP.E01/vol_vols/Users/phantom/Desktop
Element.Jsk				2022-04-01 22:15:32 UTC	2022-04-01 22:15:32 UTC	2024-12-27 00:00:51 UTC	2022-04-01 22:15:32 UTC	1391	Allocated	Allocated	unknown	/img_BelkaCTF_6_CASE240403_LAPTOP.E01/vol_vols/Users/phantom/Desktop
Prodeur.exe				2024-02-02 22:26:53 UTC	2024-02-02 22:27:14 UTC	2024-12-27 00:07:15 UTC	2024-04-02 10:15:35 UTC	6886720	Allocated	Allocated	unknown	/img_BelkaCTF_6_CASE240403_LAPTOP.E01/vol_vols/Users/phantom/Desktop

Item	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Metadata									
Name:	/img_BelkaCTF_6_CASE240403_LAPTOP.E01/vol_vols/Users/phantom/Desktop/Prodeur.exe								
Type:	File System								
MIME Type:	application/x-dosexec								
Size:	6886720								
File Name Allocation:	Allocated								
Metadata Allocation:	Allocated								
Modified:	2024-02-02 22:26:53 UTC								
Accessed:	2024-12-27 00:07:15 UTC								
Created:	2024-04-02 10:15:35 UTC								
Changed:	2024-02-02 22:27:14 UTC								
MD5:	a5a5587fced8b0d6d5d7549309621f								
SHA-256:	0ee7490fedb9247af9e5ff64042f3714								
Hash Lookup Results:	UNKNOWN								
Internal ID:	40236								

Q6: Locate the file with the MD5 hash value “0ee7490fedb9247af9e5ff64042f3714”. Can you determine the amount in euros (EUR) that the user appears to have paid for Airbnb services?

Answer:

Using Autopsy's "**File Search by Attributes**" tool and searching based on the above MD5 hash, the image file `Capture2.png` was found. Examining the content of the image (in the "Application" tab), it was determined that it is a screenshot from a transaction history. In it, a payment to Airbnb on March 26, 2024 is recorded. The amount listed is **PAB 47.3** (Balboa Panama). Converting this

amount into Euros (from Revolut), it appears that the user paid about **41.10 Euros**.

File Search by Attributes

☐ Name:

*Note: Name match is case insensitive and matches any part of the file name (not including parent path). Regular expressions are not currently supported

☐ Date: ... to ...

*Empty fields mean "No Limit" *The date format is mm/dd/yyyy

Timezone:

☒ Modified ☒ Accessed ☒ Created ☒ Changed

☐ Size:

☐ Known Status:

☒ Unknown ☒ Known (NSRL or other) ☒ Notable

☐ MIME Type:

application/andrew-inset
application/applefile
application/appxware
application/atom+xml

*Note: Multiple MIME types can be selected

☒ MD5:

☐ SHA-256:

☐ Data Source:

*Note: Multiple data sources can be selected

☐ Deleted

Listing: File Search Results 1 x

Filesystem Search Results: 1 Results

Table Thumbnail

Save Table as CSV

Name	S	C	D	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
Capture.png				2024-03-28 16:58:19 UTC	2024-04-04 11:14:02 UTC	2024-03-28 16:58:19 UTC	2024-03-01 15:46:40 UTC	407336	Allocated	Allocated	Unknown	/img/BelkaCTF_6_CASE240405_LAPTOP

File Metadata: OS Account: Data Artifacts: Analysis Results: Context: Annotations: Other Occurrences:

LAARI ADA St. Louis
Mar 27, 2024
-33.25 PAB

AIRBNB 14158800005959
Mar 26, 2024
-47.3 PAB

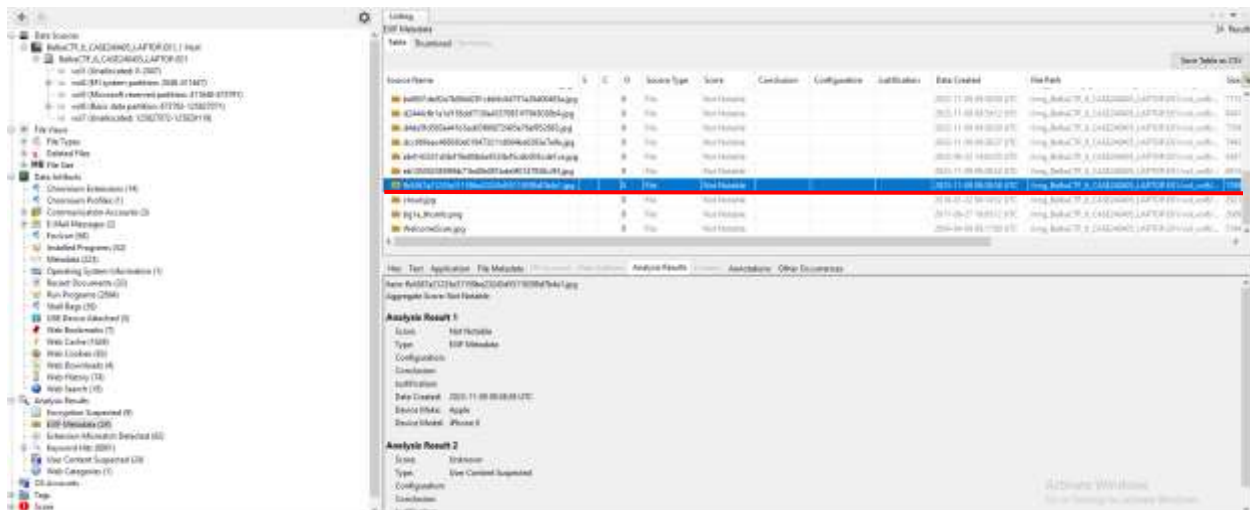
Q7: Locate and extract (Using right-click-> *Extract file(s)* option) the picture "fb4367a21233e31159b_e23245d55119298d7b4e1.JPG". Use exiftool to analyze the EXIF metadata of this picture and answer the following questions (exiftool available here:

<https://exiftool.org/>) Hint: Drag and drop the file onto the `exiftool(-k).exe` to reveal its EXIF metadata.

- What was the **Camera Model Name** of the device that took this picture?
- What was the **Orientation** of the device when it took this picture?
- What was the **Lens Model** of the device that took this picture?
- When this picture was **Originally Created** (before being moved to the computer- Provide the timestamps in local time)?

Answer:

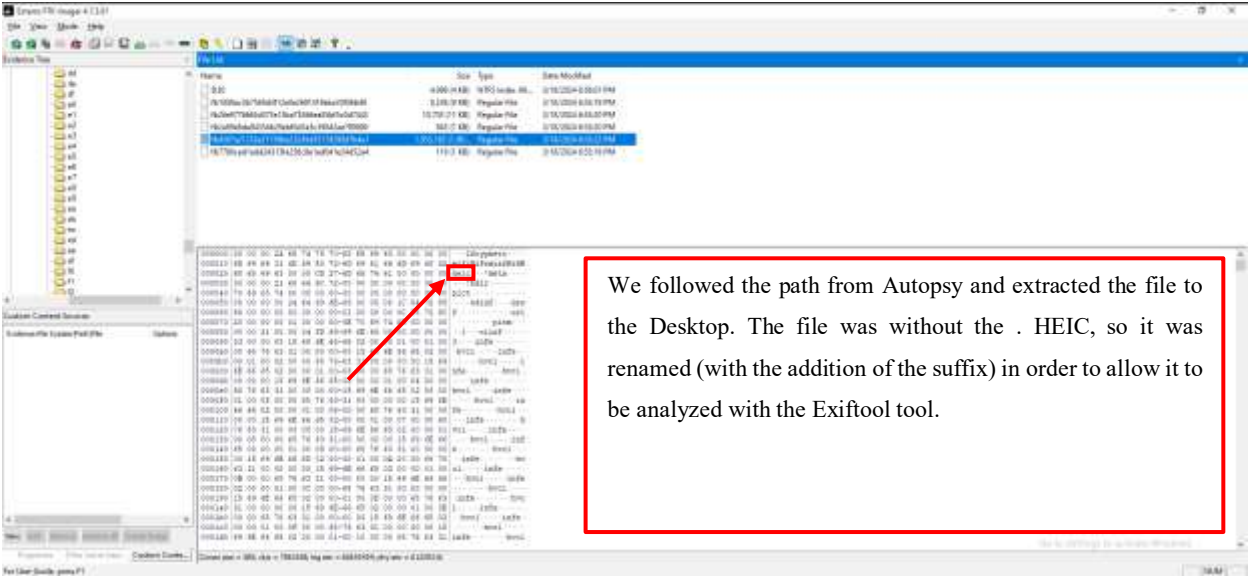
To solve question 7, it was necessary to locate and analyze the **EXIF metadata** of the «fb4367a21233e31159be23245d55119298d7b4e1.JPG file." During the process of exporting the file through Autopsy, a technical issue was encountered, as the exported file had a size of **0 bytes**, making it impossible to process it further.



```
C:\Users\Client_Win10\Desktop\exiftool-13.53_64\exiftool(-k).exe

ExifTool Version Number      : 13.53
File Name                    : fb4367a21233e31159be23245d55119298d7b4e1.jpg
Directory                    : C:/Users/Client_Win10/Desktop/exiftool-13.53_64
File Size                    : 0 bytes
File Modification Date/Time   : 2026:03:29 02:17:38+02:00
File Access Date/Time        : 2026:03:29 02:31:33+02:00
File Creation Date/Time      : 2026:03:29 02:17:38+02:00
File Permissions              : -rw-rw-rw-
Error                        : File is empty
-- press ENTER --
```

This difficulty was due to the fact that Autopsy was trying to retrieve a JPG preview from a previous resolution folder (ModuleOutput), which was no longer available in the current environment. Through hexadecimal analysis (Hex view) in the alternative FTK Imager tool, it was determined from the signature of the file that the original file was in fact in HEIC (High Efficiency Image Format) format. To circumvent this error, FTK Imager was used to successfully extract the integer file of approximately 1.9 MB in size from the forensic image. The file was then renamed with the correct ending (.heic) and parsed with the ExifTool tool.



C:\Users\Client_Win10\Desktop\exiftool-13.53_64\exiftool(-k).exe

```
ExifTool Version Number      : 13.53
File Name                    : fb4367a21233e31159be23245d55119298d7b4e1.heic
Directory                   : C:/Users/Client_Win10/Desktop/exiftool-13.53_64
File Size                    : 1953 kB
File Modification Date/Time  : 2024:03:18 20:53:22+02:00
File Access Date/Time       : 2026:03:29 04:07:36+03:00
File Creation Date/Time     : 2024:03:18 20:53:12+02:00
File Permissions             : -rw-rw-rw-
File Type                   : HEIC
File Type Extension         : heic
MIME Type                   : image/heic
Major Brand                 : High Efficiency Image Format HEVC still image (.HEIC)
Minor Version               : 0.0.0
Compatible Brands           : mif1, MiPr, miaf, MiHB, heic
Handler Type                : Picture
Primary Item Reference       : 49
Meta Image Size             : 4032x3024
Exif Byte Order              : Big-endian (Motorola, MM)
Make                        : Apple
Camera Model Name           : iPhone 8
Orientation                 : Rotate 90 CW
X Resolution                 : 72
Y Resolution                 : 72
Resolution Unit              : inches
Software                     : 16.3
Modify Date                  : 2023:11:09 09:38:58
Host Computer                : iPhone 8
Exposure Time                : 1/50
F Number                     : 1.8
Exposure Program             : Program AE
ISO                           : 160
Exif Version                 : 0232
Date/Time Original           : 2023:11:09 09:38:58
Create Date                  : 2023:11:09 09:38:58
Offset Time                  : -06:00
Offset Time Original         : -06:00
Offset Time Digitized        : -06:00
Shutter Speed Value          : 1/50
Aperture Value               : 1.8
Brightness Value             : 2.417279939
Exposure Compensation        : 0
Metering Mode                : Multi-segment
Flash                        : Off, Did not fire
Focal Length                 : 4.0 mm
Subject Area                 : 2015 1511 2217 1330
Maker Note Version           : 14
Run Time Flags               : Valid
Run Time Value               : 2805537796500
Run Time Scale               : 1000000000
Run Time Epoch               : 0
AE Stable                    : No
AE Target                    : 195
AE Average                   : 205
```



```

Focal Length In 35mm Format : 28 mm
Lens Info : 3.99mm f/1.8
Lens Make : Apple
Lens Model : iPhone 8 back camera 3.99mm f/1.8
Profile CMM Type : Apple Computer Inc.
Profile Version : 4.0.0
Profile Class : Display Device Profile

```

- The camera model is **iPhone 8**.
- The orientation of the device when shooting was **Rotate 90 CW** (90-degree clockwise rotation).
- The lens model is **iPhone 8 rear camera 3.99mm f/1.8**.
- Photo originally created at **2023:11:09 09:38:58** (local device time).

Ultimately, the combined use of **FTK Imager** and **ExifTool** tools was essential for the successful completion of the analysis, allowing for the recovery of the data despite the limitations presented by the original software.

Q8: When was the file “Spending.xlsx” last opened by the user “phorger”? (If you used Autopsy to find the answer add a ‘*Bookmark*’ tag to the artifact that supports your answer-Provide the timestamps in UTC).

Answer:

File «**Spending.xlsx**” was last opened by user “**phorger**” on: **2024-12-24 22:57:58 UTC**. It was found in subcategory “Recent Documents”.

The screenshot shows the Autopsy 4.19.0 interface. The left sidebar displays a tree view of data sources and artifacts. The main window shows the 'Recent Documents' table, which lists files accessed recently. The file 'Spending.xlsx' is highlighted in blue, indicating it is the selected artifact. Below the table, the 'Details' pane shows the file's metadata, including its path and the user who last accessed it.

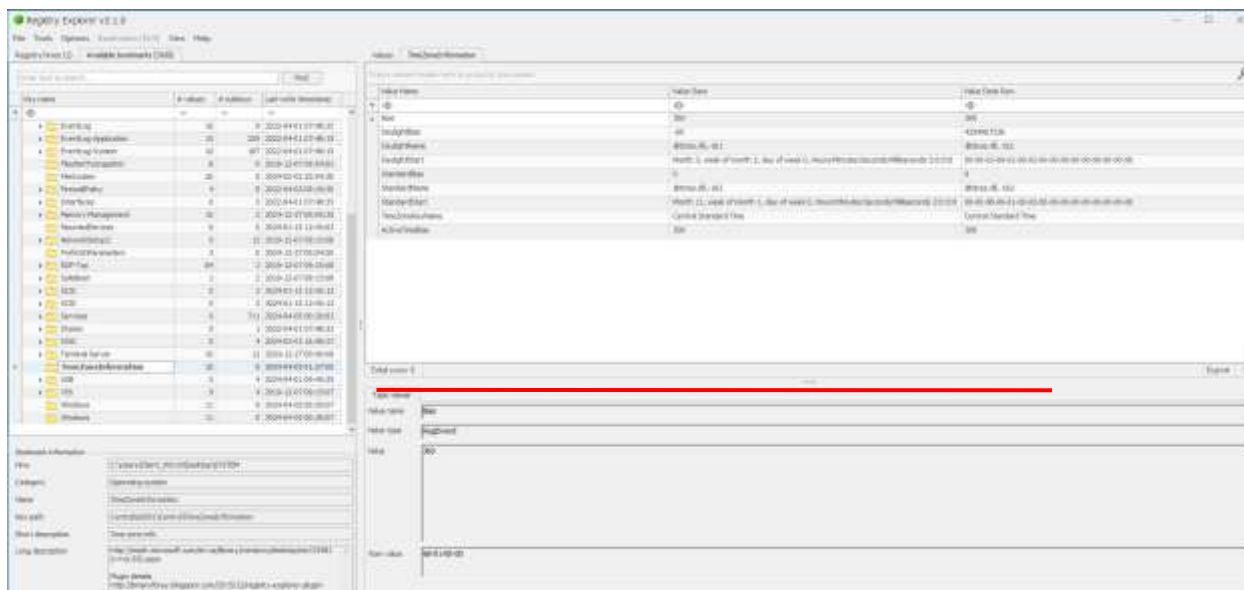
Source Name	Path	Date Accessed	Data Source	Comment
Java	%	2024-05-01 00:00:00	BulkCTF_8_CASE240405_LAPTOP.001	
CaptureLink	C:\Users\phorger\Pictures\Capture.PNG	2024-01-25 17:25:37 UTC	BulkCTF_8_CASE240405_LAPTOP.001	
MusicLink	C:\Users\phorger\Music	2024-09-09 00:00:00	BulkCTF_8_CASE240405_LAPTOP.001	
VideosLink	C:\Users\phorger\Videos	2024-09-09 00:00:00	BulkCTF_8_CASE240405_LAPTOP.001	
DesktopLink	C:\Users\phorger\Desktop	2024-09-09 00:00:00	BulkCTF_8_CASE240405_LAPTOP.001	
PicturesLink	C:\Users\phorger\Pictures	2024-09-09 00:00:00	BulkCTF_8_CASE240405_LAPTOP.001	
DownloadsLink	C:\Users\phorger\Downloads	2024-09-09 00:00:00	BulkCTF_8_CASE240405_LAPTOP.001	
DocumentsLink	C:\Users\phorger\Documents	2024-09-09 00:00:00	BulkCTF_8_CASE240405_LAPTOP.001	
775AR-04LPDF.pdf	%\775AR-04LPDF.pdf	2024-09-15 13:00:59 UTC	BulkCTF_8_CASE240405_LAPTOP.001	
Spending.xlsx	%\Spending.xlsx	2024-12-24 22:57:58 UTC	BulkCTF_8_CASE240405_LAPTOP.001	

Type	Value	SourceID
Path	%\Spending.xlsx	RecentActivity
Path ID	-1	RecentActivity
Date Accessed	2024-12-24 22:57:58 UTC	RecentActivity
Source File Path	C:\Users\phorger\Documents\Spending.xlsx	RecentActivity
Artifact ID	9233702685475776	

Q9: What was the selected time zone of this system?

Answer:

According to the analysis of the SYSTEM file (TimeZoneInformation) through the tool **Registry Explorer**, the selected system time zone was **Central Standard Time (CST)**. The exact determination of the time zone through the Registry of the system in question is of utmost importance, as it allows the correct correlation of events and the creation of a reliable timeline. While analysis tools such as Autopsy may display timestamps based on the local time of the analyzer's computer - as was the case in this case with the display of EET (Eastern European Time) readings, due to the influence of our own workstation settings - exporting the time zone directly from the SYSTEM hive reveals the actual setting of the machine. Knowing now as a geographical fact that the CST zone corresponds to a constant UTC -6 hours, the accurate calculation of the user's local time is ensured. This conversion is absolutely critical for cross-referencing actions, such as the correlation between the creation of the photo (which was recorded in the metadata in local time) and the opening of the «Spending.xlsx» file (which was recorded in Windows artifacts in absolute UTC time), thus preventing any erroneous conclusions during the examination.



Q10: Which the 'Enabled Modules' of this case were when it was analyzed by me?

Answer:

The "Ingest History" section with the "Enabled Modules" shows us which Autopsy tools were activated to search the disk. Mentioning these in the final Report is very important because if another expert needs to check the case, he must know exactly what tools we used. Only in this way will it be able to repeat the same steps and confirm our findings. In this case, the successful extraction of the conclusions in the previous questions was directly based on the following selected modules:

- **Recent Activity:** This module was solely responsible for retrieving the browsing history in Microsoft Edge (Query 4) as well as locating the shortcut file «Spending.xlsx.lnk" in "Recent Documents" (Query 8).
- **Embedded File Extractor:** Allowed Autopsy to "open" the compressed files in the background, giving us direct access to the Powder.exe inside the powder.zip file (Question 5).
- **Hash Lookup:** It was the necessary function that allowed the instant detection of the «Capture2.png" image through the database, searching exclusively with its MD5 hash (Query 6).

That list therefore demonstrates which categories of 'Data Artifacts' have been made available for examination, defining the scope and potential of the present investigation.

Ingest History:

Job 1:

Data Source: BelkaCTF_6_CASE240405_LAPTOP.E01

Status: COMPLETED

Enabled Modules:

- Recent Activity
- Hash Lookup
- File Type Identification
- Extension Mismatch Detector
- Embedded File Extractor
- Picture Analyzer
- Keyword Search
- Email Parser
- Encryption Detection
- Interesting Files Identifier
- Central Repository
- PhotoRec Carver
- Virtual Machine Extractor
- Data Source Integrity
- Android Analyzer (aLEAPP)
- DJI Drone Analyzer
- YARA Analyzer
- iOS Analyzer (iLEAPP)
- GPX Parser
- Android Analyzer

Epilogue

This paper highlighted in practice the complexity and methodicality required in Digital Forensics. Through the use of specialized tools such as Autopsy, FTK Imager, and Registry Explorer, it became possible to recover critical digital evidence from a disk image (. E01). During the analysis, it was confirmed that automated data extraction through Ingest Modules (such as Recent Activity or Keyword Search) rapidly accelerates research. However, as demonstrated by analyzing the HEIC photo (Query 7) and identifying the system's actual time zone (Query 9), automated tools may exhibit errors or misleading readings. In these cases, critical thinking by the analyst, manual analysis (e.g., Hex view), and deep knowledge of operating system structures (such as Windows Registry) are the only way to draw the right conclusion. Finally, the creation of the final report (Forensic Report) completed the research, translating the technical findings into a structured, legally documented and reproducible deliverable, ready to be presented to any interested party.